

## 2.3 S3 Storage Management

There are different store management operations that are available as a part of S3 which can help in cost optimisation, analytics and cleanup of data. If you click on Management tab in S3 bucket, it looks something like this

Let's look at some of these options:

1. **Lifecycle** : S3 was designed primarily for rapid retrieval of data. But this fast retrieval comes with a significant cost and if the data becomes huge, it can be pretty expensive. Amazon has a solution for that too. You can create various life-cycle rules to move the data out of S3. If your data is not relevant to you after a certain time period, you can add an expiration rule and S3 will clean up the particular folder for you after the data becomes older than a certain threshold. If you don't want to permanently delete the data, another option is moving the data to Glacier. Glacier is a very cheap data store which costs as little as \$0.01 per GB per month where you can put the old data of S3 but it takes 3-5 hours to retrieve the data.
2. **Replication** : You can add cross region replication rules in case you want to have faster data access across multiple regions.
3. **Analytics** : This Amazon S3 analytics feature observes data access patterns and helps you determine when to transition from the less frequently accessed STANDARD storage to the STANDARD\_IA (IA, for infrequent access) storage class. You can set it up on the entire bucket or a prefix path inside the bucket.
4. **Metrics** : Normally, AWS provides you with storage metrics once per day, free of cost but if you want to opt for request metrics per minute you can opt for this paid metrics option.

## 3. EC2 & S3

Often, you will have a situation where you would need to establish communication between various AWS services and one such common scenario will be communication between EC2 and S3. You have learned EC2 and S3 in the above sections and after having a hands-on tutorial, you should now have become quite familiar with them. You have also learned how to make use of AWS CLI to exchange the files between S3 bucket and local machine. Similarly, EC2 is like a local machine but it is running on the cloud. By doing SSH into the EC2 machine you can use the same strategies that you have learned in section 2.2.2 of this tutorial to establish communication between S3 and EC2 and transfer whatever data you need from S3 to EC2 or vice-versa. 